Mingyu "Myles" Ouyang

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Research Focus Areas

My research focuses on digital (computer-use/game) agents and the interaction and productivity gains they enable. I currently focus on game environments with rich visual observations, long-horizon goals, and dynamic mechanics. I aim to improve agents' perception, reasoning, and control for their future embodiment.

Education

National University of Singapore — Singapore
Ph.D. in Computer Science — *Jan. 2025 – Present*
Supervisors: Prof. Hwee Tou Ng and Prof. Mike Zheng Shou

National University of Singapore — Singapore
M.Sc. in Data Science and Machine Learning — *Aug. 2023 – Dec. 2024*

Wuhan University — Wuhan, China
B.Sc. in Geospatial Informatics and Digitalized Technology — *Sep. 2019 – Jul. 2023*

Publications

(* equal contribution; † corresponding author)

[Under review, 2026.10] Player-1: Learning to Play from Synthetic Experience
Mingyu Ouyang, Yiran Wang, Siyuan Hu, Xingyilang Yin, Huiqiang Sun, Guangzhi Wang, Junfu Pu, Cheng Cheng, Honghao Chen, Ying Shan, Mike Zheng Shou†, Hwee Tou Ng†
- Synthetic environments improve transferable agent capabilities: An agentic post-training framework that learns general game control beyond supervised demonstrations.

[ECCV '26] GameWorld: Towards Standardized and Verifiable Evaluation of Multimodal Game Agents
Mingyu Ouyang*, Siyuan Hu*, Kevin Qinghong Lin, Hwee Tou Ng†, Mike Zheng Shou†
- Evaluates computer-use and general-purpose game agents with 34 games and 170 tasks.
- Received the Best Paper Runner-Up Award at the Multimodal Digital Agents Workshop, ECCV 2026.
- Adopted for multimodal agent evaluation in the Seed-2.1 model card from ByteDance Seed and VISTA from MIT.

[CVPR '26] FocusUI: Efficient UI Grounding via Position-Preserving Visual Token Selection
Mingyu Ouyang, Kevin Qinghong Lin, Mike Zheng Shou†, Hwee Tou Ng†
- Selects instruction-relevant visual tokens for low-cost, efficient UI grounding.
- Received the Compute Transparency Champion Award at CVPR 2026.

[AAAI '26] SlideTailor: Personalized Presentation Slide Generation for Scientific Papers
Wenzheng Zeng*, **Mingyu Ouyang***, Langyuan Cui*, Hwee Tou Ng†
- Generates preference-aligned presentation slides for scientific papers.

[arXiv '24] The Dawn of GUI Agent: A Preliminary Case Study with Claude 3.5 Computer Use
Siyuan Hu, **Mingyu Ouyang**, Difei Gao, Mike Zheng Shou†
- An out-of-the-box framework for desktop GUI agents, supporting multiple operating systems and models.

[CVPR '24] AssistGUI: Task-Oriented PC Graphical User Interface Automation
Difei Gao, Lei Ji, Zechen Bai, **Mingyu Ouyang**, Peiran Li, Dongxing Mao, Qinchen Wu, Weichen Zhang, Peiyi Wang, Xiangwu Guo, Hengxu Wang, Luowei Zhou, Mike Zheng Shou†
- A desktop GUI agent framework for task decomposition, GUI parsing, action generation, and reflection.

[TIP '24] JPEG Quantized Coefficient Recovery via DCT Domain Spatial-Frequential Transformer
Mingyu Ouyang, Zhenzhong Chen†
- Recovers JPEG-quantized DCT coefficients with spatial-frequency modeling across compression quality factors.

Professional Experience

ARC Lab, Tencent IEG
Research Intern (Project Up) — *2026 – Present*
- Research on multimodal digital agents for game environments and agent-environment interaction.

UseIt AI Singapore

Co-founder *Mar. 2025 – Oct. 2025*

- Led the GUI agent development and workflow orchestration, focusing on creating, editing, and deploying personal agents.

SenseTime Research Singapore Singapore

Research Intern *Dec. 2023 – Oct. 2024*

- Developed efficient low-level deep learning models for RAW image processing, including noise synthesis and learned image signal processing for built-in camera and smartphone modules.

Tencent Map, Tencent Beijing, China

Algorithm Development Intern *Mar. 2022 – Oct. 2022*

- Implemented GPS data processing using Hadoop and Spark. Developed and maintained an internal service that automatically executes daily tasks, handling over 10 billion callbacks per hour.

Springturn Education Technology Zhuhai, China

Co-founder, Software Developer *Nov. 2020 – Sep. 2021*

- Co-founded the company and secured angel investment exceeding ¥1 million in the second year of business.

Research Experience

Show Lab, NUS Aug. 2023 – Present

- Research on multimodal digital agents and agent-environment interaction, with a current focus on game environments.

IIP Lab, Wuhan University Oct. 2022 – Jul. 2023

- Research focus: Low-level vision tasks, including image super-resolution, denoising, JPEG artifact removal.

State Key Lab of Info. Engineering in Surveying, Mapping and Remote Sensing Sep. 2021 – Jun. 2022

- Research focus: 3D scene perception, fusing point clouds and BEV images for road-scene understanding.

Open-Source Projects

[GitHub 2k stars] Computer-Use OOTB: showlab/computer_use_ootb

- Core developer and maintainer. A desktop GUI agent framework for Windows/macOS with API-based or self-hosted model and local execution.

[GitHub 1.9k stars] ShowUI: showlab/showui

- Maintainer. A training and inference codebase for a lightweight (2B) VLA model for GUI agents.

Miscellaneous

Wuhan University Football Team

- Served as captain for three years and led the team to win the championship at the 16th Hubei Province Games in 2022.

Technical Skills

- **Programming Languages:** Fluent with agentic coding tools (Codex, CC).
- **AI/ML Frameworks:** PyTorch, Transformers, vLLM, VeRL.
- **Computer-Use Agent Infrastructure:** On-device control (Windows, macOS), cloud services (AWS, Azure), agent framework orchestration, and browser-based roll-out infrastructure.